



The Mid-Atlantic Ridge

Part of a global oceanic ridge system, a huge underwater mountain chain that is pushing continents apart

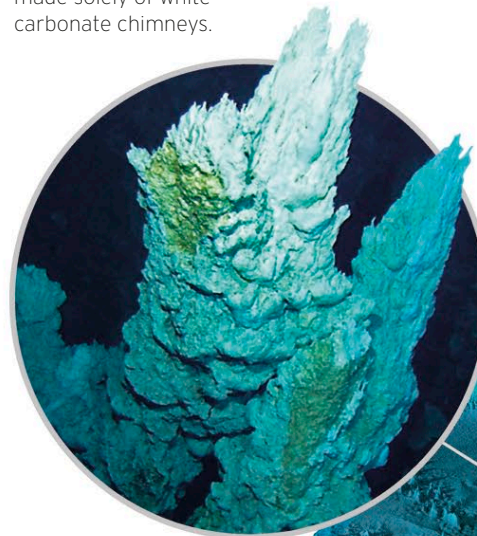
Running down the middle of the Atlantic Ocean, the Mid-Atlantic Ridge is a slow-spreading submarine ridge system that extends around 10,000 miles (16,000 km) from the Arctic to beyond the southern tip of Africa. The ridge marks a boundary where the North and South American tectonic plates are diverging from the Eurasian and African Plates.

Mountains and valleys

The path of the ridge mirrors the contours of the continents it not only divides but is also pushing apart at a rate of $\frac{3}{4}$ –2 in (2–5 cm) per year—much slower than the $\frac{6}{4}$ in (16 cm) per year of the East Pacific Rise (see pp.322–23). The Mid-Atlantic Ridge rises $1\frac{1}{4}$ –2 miles (2–3 km) above the seafloor in a long chain of underwater mountains, and it has a deep central rift valley along its crest. Where the ridge has pushed above sea level, it has created landmasses such as the volcanic islands of Iceland and the Azores.

▽ CARBONATE TOWERS

During 2000, an undersea research team filmed a cluster of ghostly spires 90–200 ft (27–61 m) high. This is the Lost City, the only known hydrothermal field made solely of white carbonate chimneys.



► DIVERGENT BOUNDARY

Mid-oceanic ridges such as the Mid-Atlantic Ridge mark areas where tectonic plates diverge as new crust forms. Deep below the ocean surface, the upwelling of magma from the mantle causes volcanic eruptions, creating mountain ranges, rift valleys, and hydrothermal vents along the ridge's length.

Transform fault

Instead of forming a straight line, the ridge follows a stepped path, with sections offset along faults called transform faults. Movement along these faults often sets off shallow earthquakes.

South America

continental crust



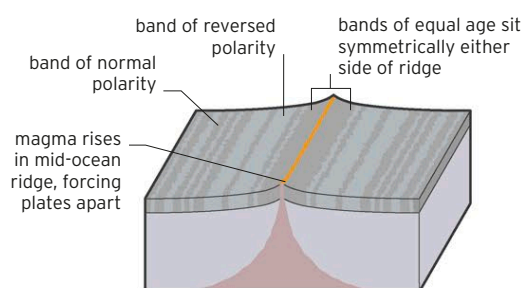
△ SEABED PILLOWS

Lava erupted along the ridge cools rapidly in the cold seawater to form mounds 20 in–3 ft (0.5–1 m) across called pillow lavas.

North America

MAGNETIC BANDS AT SPREADING RIDGES

Earth's magnetic field periodically reverses, changing the north magnetic pole to the south magnetic pole, and vice versa. Why this occurs is unknown, but it is recorded in bands of basaltic rock on either side of mid-ocean ridges. Metallic elements within new crust created at a spreading ridge are aligned with Earth's polarity at the time the basaltic lava emerges. The study of these bands has been a crucial line of evidence in support of the theory of plate tectonics because it helped prove that Earth's plates move.



westward movement of South American Plate

oceanic crust

Asthenosphere

Below the lithosphere lies the asthenosphere, part of Earth's mantle. Because it remains hot and fluid, it acts like a lubricant on the underside of tectonic plates, allowing them to move.